

Delayed Start and Isolation of Superunit Birth Events

Finite-Prefix Invariance versus Recurrent Viability Obstructions

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Abstract

RH-144 proves that two coarse finite chains have empty controlled viability kernels because every candidate control has a zero-state floor above one. We determine what this means asymptotically. Finite deletion does not change

$$\limsup y_n, \quad \liminf a_n, \quad \liminf (1 - \sqrt{y_n})_+^4 a_n.$$

Hence a proof may start after a finite exceptional prefix, provided a reset enters a future controlled-invariant tube. Conversely, if

$$m_n = \inf_{c \in \mathcal{C}_n} F_{n,c}(0) \geq 1$$

for infinitely many n , then every nonnegative controlled trajectory has $\limsup y_n \geq 1$. A recurrent superunit floor is therefore a genuine asymptotic obstruction; a finite one is not.

The finite archive contains exactly two superunit birth coefficients, both at the left $\sigma = 0.08$ anchor for thresholds 10^{-8} and 10^{-6} . Their full candidate-family minimum floor is $9.535 > 1$, and both backward kernels are empty. Nevertheless the cofinal anchor suffix beginning at $\sigma = 0.04$ has 18/18 forward-safe and outward-positive chains. Its minimum terminal directional floor exceeds 1.5×10^{-10} ; at the two finest anchors the minimum exceeds 9.7×10^{-4} . This rigorously isolates the known failures as finite-prefix events. It does not prove that no analogous birth occurs at an untested future level.

1 Finite failures and asymptotic statements

The RH-139 frontier is eventual: it asks for $\limsup y_n < 1$ and $\liminf a_n > 0$. Such conditions deliberately ignore finitely many initial indices. A finite envelope failure can invalidate a certificate that begins at level zero while leaving an eventual theorem logically possible.

This observation is not permission to erase an inconvenient point from one continuous trajectory. A delayed start must specify a later state enclosure or reset and prove the future recurrence from there. RH-144 supplies the right language: the reset state must enter a positive backward/block kernel [Aubin, 1991].

2 Cofinal invariance and delayed support

We record the elementary but essential theorem explicitly.

Theorem 1 (Finite-prefix invariance). *For any real sequence z_n and integer N ,*

$$\limsup_{n \rightarrow \infty} z_n = \limsup_{n \rightarrow \infty} z_{n+N}, \quad \liminf_{n \rightarrow \infty} z_n = \liminf_{n \rightarrow \infty} z_{n+N}.$$

Consequently the RH-139 hypotheses and directional lower are unchanged by deleting a finite prefix.

Proof. The supremum and infimum of a tail are unchanged after removing finitely many terms once the tail index passes those terms. Taking limits proves both identities. Apply them to y_n , a_n , and the directional candidate. $\square$

Corollary 2 (Delayed-start support). *Suppose that at some level N a validated reset gives $y_N \leq r$, every future controlled block preserves a tube with partial-block upper $\bar{y} < 1$, and $\liminf_{n \geq N} a_n \geq a_* > 0$. Then*

$$\liminf_{n \rightarrow \infty} B_n \geq (1 - \sqrt{\bar{y}})^4 a_* > 0,$$

regardless of the finite prefix before N .

Proof. The block policy gives the tail limsup bound from N onward. Apply the RH-139 support theorem to the shifted sequence and then Theorem 1. $\square$

3 The recurrent-floor obstruction

Finite deletion has a sharp opposite.

Theorem 3 (Infinitely recurrent superunit floors). *Let every controlled envelope be increasing on $[0, \infty)$ and define*

$$m_n = \inf_{c \in \mathcal{C}_n} F_{n,c}(0).$$

If $m_n \geq 1$ for infinitely many n , then every nonnegative controlled trajectory obeys $\limsup y_n \geq 1$. In particular the RH-139 tail packet is impossible for that control family.

Proof. At every such index and for every control, $y_{n+1} = F_{n,c_n}(y_n) \geq F_{n,c_n}(0) \geq m_n \geq 1$. This occurs infinitely often, so the upper limit is at least one. $\square$

The theorem is architecture relative: enlarging the control family can lower m_n . Within a fixed family it is exact. It also shows why an all-level proof must exclude recurrent bad births, not merely show that their density tends to zero.

start cutoff	suffix chains	forward safe	outward positive	minimum positive terminal floor
0.16	30	28	28	1.04×10^{-10}
0.08	24	22	22	1.55×10^{-10}
0.04	18	18	18	1.55×10^{-10}
0.02	12	12	12	9.76×10^{-4}
0.01	6	6	6	2.14×10^{-3}

Table 1: Cofinal finite-anchor suffixes.

4 Finite suffix audit

The RH-137 archive has 330 transitions. Exactly two selected birth coefficients exceed one:

$$q = 9.405889994899066.$$

Both occur on the left $\sigma = 0.08$ anchor, at the final transition, for thresholds 10^{-8} and 10^{-6} . RH-144 strengthens this observation by minimizing over the entire gauge family: the least zero-state floor is still $9.5352 > 1$, and the predecessor kernel is empty in both cases.

We order the five finite anchors toward finer scale and delete successively larger prefixes. The same audit is applied to the forward RH-137 chains and the outward RH-138 directional composition.

Thus $\sigma = 0.04$ is the first completely clean archived suffix. The coarser $\sigma = 0.16$ chains happen to be safe, but a cofinal start after the bad 0.08 anchor necessarily discards them as part of the finite prefix.

5 Consequence and claim boundary

The known coarse failures no longer need to be “repaired” in an eventual theorem. A valid route may start after them. Its remaining obligations are precise: produce a validated reset at the delayed start, prove a common future block kernel and partial safety gap, and prove the normalized-base liminf.

The clean 18-chain suffix is finite evidence for this strategy, not proof that every future level is clean. We have not excluded recurrent superunit births analytically, constructed an all-level delayed reset, proved a uniform tail gap or base liminf, established Stage A, constructed a Hilbert–Polya operator, identified zeta zeros, or proved the Riemann Hypothesis.

References

Jean-Pierre Aubin. *Viability Theory*. Birkhäuser, 1991.

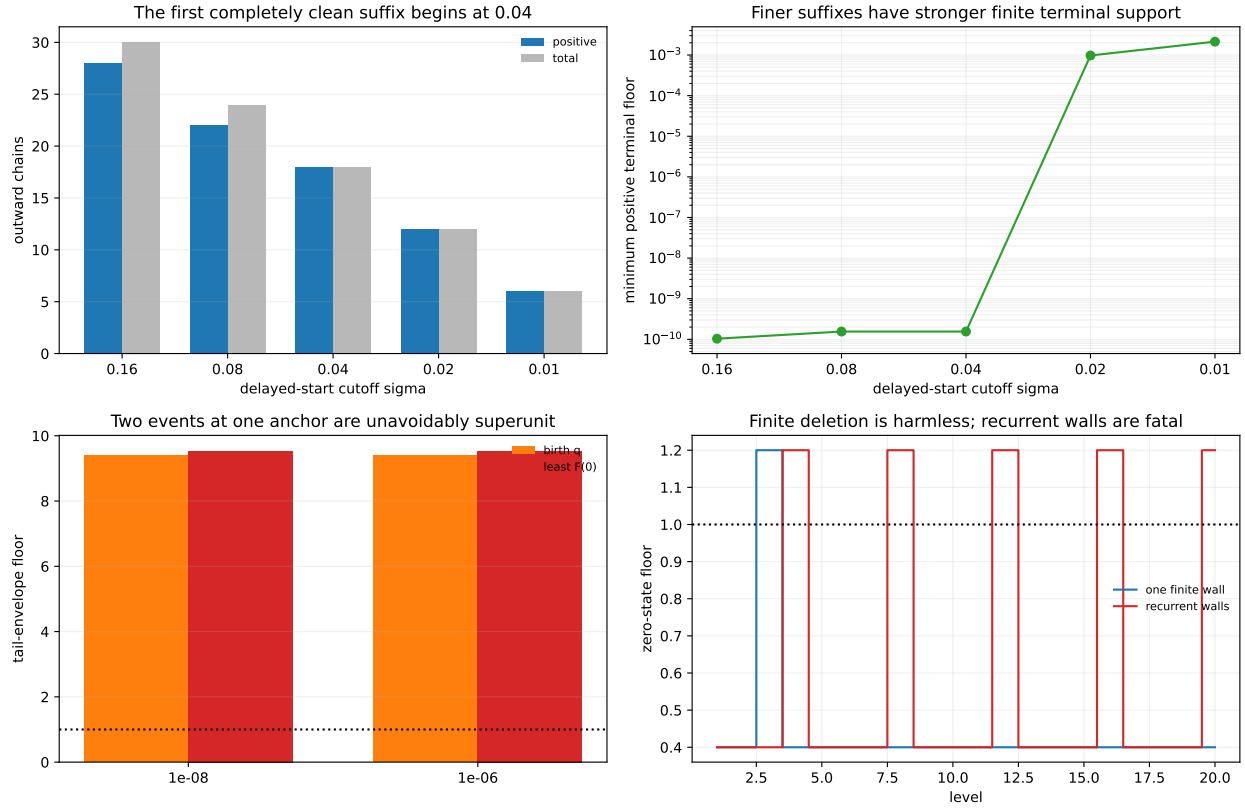


Figure 1: Suffix classifications and floors, the two isolated superunit events, and the finite-versus-recurrent distinction.